

Group: 3

UI&UX Development

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Project 2: AI Auto-Guide for Chronic Illness: Food & Vitamin Safety

Project Idea: This project is a website that automatically creates custom nutritional guides for people with chronic illnesses. Instead of searching, the user just selects their health condition, and the AI instantly generates a full list of **Foods and Vitamins** that are safe or dangerous for them. Users can save these lists to their "Favorites" to use as a quick reference and find **partner pharmacies** that have the required vitamins in stock.

Key Features (What the user gets):

1. **Automatic Food & Vitamin Lists:** Once the user selects their illness, the website builds a complete guide. It shows "**What to Eat/Take**" and "**What to Avoid**" for both food and supplements.
2. **Safety Explanations:** The AI explains *why* a certain vitamin or food is on the list (for example: "*Avoid Vitamin C supplements: Large doses can be risky for your specific kidney condition*").
3. **Favorites & Saving:** Users can "Star" their personalized results. This saves the list to their profile so they can check it quickly on their phone at the pharmacy or grocery store.
4. **Combined Health Profiles:** If a user has more than one illness, the AI combines the medical rules for both to make sure the vitamins and foods are safe for their unique situation.
5. **Condition Explanation:** Along with the medicine suggestion, the website briefly explains what the identified condition is in simple, non-technical language so the user understands what they're dealing with.
6. **Arabic & English Support:** The full experience — photo results, medicine names, pharmacy info, and chat — works in both Arabic and English.

7. **Natural Follow-Up Chat (chatbot):** After the result appears, the user can ask questions like *"Does this cream have side effects?"* or *"Can I use this while pregnant?"* and get an instant AI answer — no manual content writing needed.

"Pro" Features:

1. **Smart Health Updates:** Users can upload new medical reports or lab results. The AI reads these files and automatically updates the food and vitamin lists to match the user's current health.
2. **Doctor Connection:** Pro users can optionally share their updated health profile with a real doctor. The doctor can review the AI-generated lists and send feedback or changes through the app.
3. **Dynamic Updates:** Whenever new health information is added, the AI instantly recalculates the "Safe" and "Avoid" lists to keep them accurate in real-time.

Key Features (Future Work):

Dealing with user suffering of skin problem :

The Goal: when upload a photo of a skin problem (like a rash or pimple), and website tells user:

1. **Skin Photo Analysis:** The user uploads or takes a live photo of their skin condition (rash, pimple, dryness, etc.) and the AI instantly identifies what it likely is — no account or sign-up needed to start.
2. **Local vs. International Medicine Suggestions:** For every identified condition, the website shows two options — an affordable Egyptian local brand (e.g. *Dermovate*, *Elocon*) and an international brand — so the user can choose based on their budget.

How the AI is Used:

We use the **Gemini 1.5 Flash** model as the intelligence behind the website. It performs three main jobs:

- **Medical Fact-Checking (RAG):** The AI doesn't "guess" the lists. We use a method called **Retrieval-Augmented Generation (RAG)** to connect the AI to trusted medical databases. This ensures the vitamin and food lists are based on real doctor-approved guidelines.

- **Reading Medical Files:** The AI uses **Optical Character Recognition (OCR)** and **Natural Language Processing (NLP)** to "read" and understand uploaded medical reports. It extracts the most important lab results to update the user's profile automatically.
- **Reasoning and Risk Finding:** The AI compares the user's illnesses and medications to find "hidden" risks—like a vitamin that might be good for your heart but dangerous for your specific blood pressure medication.

How We Make Profit (Business Model):

- **Pharmacy Partnerships:** We partner with local pharmacies to list their available vitamins. For every long-term customer that buys from them through our app, we take a percentage of the profit. This guarantees the pharmacy a loyal customer and provides us with a steady income.
- **Pro Subscriptions:** We offer a **Pro Version** of the account. Users pay a subscription fee to access advanced features like medical report scanning, automatic health updates, and doctor integration.